

A/B test analysis report: Assessing the impact of adding more relevant generic names for dishes, localized names, and alternative names for dishes.

1. Research methodology.

Data processing: The original event logs were aggregated at the unique user (user_id) level. This allowed us to create a behavior profile for each user and calculate the conversion rate from the total number of users. **Object of Analysis:** The analysis was conducted across the app's **language versions**, which aligns with the hypothesis being tested regarding the impact of localized and alternative dish names on user experience.

2. Instrumentation Used.

The Z-test and Fisher's Exact Test were used to evaluate purchase conversion. The nonparametric Mann-Whitney U test was used to analyze interaction intensity (number of likes and dislikes).

3. Analysis Results by Language Segment.

Based on the presented statistical data, the following indicators were established:

English: A statistically significant positive result was found. Conversion increased from 11.78% to 12.55% ($p=0.0273$, Lift +6.57%). A significant decrease in the average number of dislikes from 0.13 to 0.11 ($p=0.0003$) was also recorded, indicating improved dish identification accuracy.

German: A statistically significant decrease in key metrics was recorded. Conversion decreased from 7.71% to 3.90% ($p=0.0385$), and the average number of likes decreased from 0.22 to 0.16 ($p=0.0122$).

Italian: Positive conversion dynamics were observed (Lift +28.73%), but with the current sample size (1,680 users), the result is not statistically significant ($p=0.2796$). Changes in the number of likes and dislikes also did not reach the significance threshold.

Dutch: The change in conversion was insignificant (Lift +19.45%, $p=0.6326$). However, a statistically significant increase in the average number of likes was recorded from 0.06 to 0.13 ($p=0.0397$), indicating increased satisfaction with the dish identification process in this segment. It is a leading indicator that could lead to increased conversion with a larger sample size.

French & Danish: These segments demonstrate a high relative increase in conversion (+39.34% and +128.57%), but due to the small data volume, the results are not statistically confirmed in a two-tailed test ($p=0.0971$ and $p=0.0518$, respectively).

4. Recommendations.

English-language segment: A full rollout of the changes is recommended, as a simultaneous improvement in business metrics and qualitative identification indicators has been recorded.

German-language segment: The test implementation should be paused and an audit of the localized names should be conducted, as the current changes have led to a significant decrease in conversion.

Italian, Dutch, French, and Danish segments: Continued data collection is recommended. Despite positive trends, the current data volume is insufficient to make a final decision. In future analyses, using proxy metrics, such as events occurring further up the funnel (or conversions calculated from trial installs), typically have a higher frequency and absolute value and allow for faster achievement of statistical significance.

Purchase conversion analysis. TOP-15 language versions by users count.

Language	Users	Users (C/T)	Purchases (C/T)	CR Control	CR Test	Rel. Lift	P-Value (Z-test)	P-Value (Fisher)	Power	Significant
English	34804	17273 / 17531	2034 / 2200	11.78%	12.55%	+6.57%	0.0273	0.0280	59.77%	Yes
Portuguese	5293	2663 / 2630	201 / 181	7.55%	6.88%	-8.82%	0.3493	0.3666	15.29%	No
French	2609	1306 / 1303	41 / 57	3.14%	4.37%	+39.34%	0.0971	0.1004	38.18%	No
Italian	1680	839 / 841	31 / 40	3.69%	4.76%	+28.73%	0.2796	0.3320	18.98%	No
Spanish	1215	571 / 644	69 / 72	12.08%	11.18%	-7.48%	0.6234	0.6541	7.09%	No
Dutch	808	411 / 397	13 / 15	3.16%	3.78%	+19.45%	0.6326	0.7024	6.92%	No
German	658	350 / 308	27 / 12	7.71%	3.90%	-49.49%	0.0385	0.0465	54.37%	Yes
Danish	334	167 / 167	7 / 16	4.19%	9.58%	+128.57%	0.0518	0.0819	49.39%	No
Swedish	248	124 / 124	4 / 7	3.23%	5.65%	+75.00%	0.3548	0.5395	15.04%	No
Russian	152	80 / 72	2 / 0	2.50%	0.00%	-100.00%	0.1768	0.4981	27.11%	No
Polish	130	68 / 62	1 / 1	1.47%	1.61%	+9.68%	0.9475	1.0000	2.91%	No
Norwegian	129	61 / 68	3 / 5	4.92%	7.35%	+49.51%	0.5670	0.7212	8.26%	No
Turkish	75	37 / 38	2 / 3	5.41%	7.89%	+46.05%	0.6657	1.0000	6.33%	No
Romanian	74	41 / 33	0 / 2	0.00%	6.06%	+0.00%	0.1100	0.1955	35.87%	No
Japanese	73	41 / 32	4 / 1	9.76%	3.12%	-67.97%	0.2657	0.3772	19.85%	No

User experience analysis (number of likes and dislikes). TOP-15 language versions by users count.

Language	Users	Users (C/T)	Avg Likes (C/T)	P-Value Likes	Avg Dislikes (C/T)	P-Value Dislikes	Significant Likes	Significant Dislikes
English	34804	17273 / 17531	0.29 / 0.27	0.0964	0.13 / 0.11	0.0003	No	Yes
Portuguese	5293	2663 / 2630	0.17 / 0.17	0.7298	0.08 / 0.09	0.9938	No	No
French	2609	1306 / 1303	0.10 / 0.12	0.3731	0.04 / 0.05	0.6472	No	No
Italian	1680	839 / 841	0.17 / 0.23	0.3454	0.05 / 0.04	0.1379	No	No
Spanish	1215	571 / 644	0.34 / 0.19	0.7086	0.12 / 0.08	0.0596	No	No
Dutch	808	411 / 397	0.06 / 0.13	0.0397	0.03 / 0.03	0.3866	Yes	No
German	658	350 / 308	0.22 / 0.16	0.0122	0.04 / 0.05	0.8540	Yes	No
Danish	334	167 / 167	0.07 / 0.14	0.3565	0.04 / 0.04	0.3183	No	No
Swedish	248	124 / 124	0.13 / 0.15	0.8203	0.03 / 0.02	0.9903	No	No
Russian	152	80 / 72	0.05 / 0.07	0.8429	0.03 / 0.00	0.1813	No	No
Polish	130	68 / 62	0.18 / 0.06	0.5456	0.06 / 0.10	0.9193	No	No
Norwegian	129	61 / 68	0.07 / 0.21	0.0673	0.02 / 0.04	0.6256	No	No
Turkish	75	37 / 38	0.11 / 0.03	0.1621	0.03 / 0.03	1.0000	No	No
Romanian	74	41 / 33	0.07 / 0.00	0.1188	0.02 / 0.03	0.8922	No	No
Japanese	73	41 / 32	0.63 / 0.00	0.0258	0.00 / 0.00	1.0000	Yes	No